

ALEXANDRIA BORG EL ARAB INTL, Egypt

WMO: 623601

Lat: 30.918N

Lon: 29.696E

Elev: 177

StdP: 14.60

Time Zone: 2.00 (E02)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.3	46.1	33.5	28.2	57.1	35.7	30.8	57.4	29.9	53.1	26.9	57.7	6.0	240	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 15.0	(c) 94.8	(d) 69.8	(e) 91.1	(f) 72.0	(g) 88.2	(h) 73.6	(i) 77.7	(j) 86.0	(k) 76.9	(l) 85.0	(m) 76.1	(n) 84.1	(o) 12.0	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	134.0	81.5	74.8	131.1	81.3	73.5	125.5	81.0	41.2	86.1	40.4	84.7	39.7	84.0	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 23.0	(b) 20.6	(c) 18.7	(e) 40.8	(f) 105.4	(g) 2.3	(h) 2.8	(i) 39.2	(j) 107.4	(k) 37.8	(l) 109.1	(m) 36.5	(n) 110.7	(o) 34.9	(p) 112.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.1	56.0	57.8	61.6	66.3	71.8	77.0	79.9	80.9	78.2	72.9	66.3	59.7
	DBStd	9.33	3.17	4.25	4.59	4.63	5.04	3.67	2.26	2.00	2.60	3.34	3.88	3.67
	HDD50	2	1	1	0	0	0	0	0	0	0	0	0	0
	HDD65	846	279	207	128	33	2	0	0	0	0	0	29	168
	CDD50	6973	187	218	361	490	677	810	926	959	846	709	488	302
	CDD65	2342	0	4	23	73	214	360	461	494	396	245	67	5
	CDH74	21140	6	66	267	694	1701	3131	4565	5275	3538	1591	279	27
CDH80	7264	0	15	97	260	670	1093	1598	1990	1148	341	47	5	
Wind	WSAvg	9.5	9.0	9.3	10.2	10.7	10.2	10.4	10.7	9.4	9.3	8.2	7.8	8.6
Precipitation	PrecAvg	6.8	1.7	1.2	0.6	0.2	0.1	0.0	0.0	0.0	0.0	0.3	0.9	1.7
	PrecMax	10.7	3.3	2.7	1.7	0.8	0.5	0.1	0.0	0.0	0.2	1.2	3.4	3.6
	PrecMin	3.5	0.7	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.4
	PrecStd	1.6	0.7	0.6	0.4	0.2	0.1	0.0	0.0	0.0	0.1	0.3	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.7	80.5	91.0	94.2	102.0	100.7	94.8	95.1	94.8	91.3	85.6	77.2
		MCWB	57.3	60.8	63.1	65.3	68.1	69.9	74.1	76.0	70.7	69.4	67.1	62.4
	2%	DB	69.4	73.8	80.7	86.3	93.0	92.9	89.8	90.0	89.9	86.1	80.2	71.7
		MCWB	57.2	58.4	61.2	64.5	66.5	71.1	75.1	76.3	73.3	69.5	66.7	60.4
	5%	DB	66.4	69.7	75.1	82.1	86.3	88.0	88.0	88.2	87.5	83.1	76.9	69.5
		MCWB	56.5	58.2	59.8	63.1	66.9	71.9	75.0	75.6	73.8	70.1	64.8	59.9
	10%	DB	64.4	66.3	70.2	76.8	82.3	85.8	86.3	87.7	85.8	80.9	74.8	67.7
		MCWB	55.3	56.6	58.7	62.4	66.6	71.8	74.4	75.4	73.1	69.3	64.0	58.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.9	65.1	66.6	69.6	72.0	76.6	78.8	79.2	78.0	74.6	71.8	65.3
		MCDB	68.1	75.5	80.1	84.9	84.7	86.1	87.9	87.8	85.8	83.3	78.5	71.4
	2%	WB	59.3	61.0	63.0	66.8	70.0	75.3	77.2	78.1	76.3	72.7	69.3	62.9
		MCDB	66.3	69.5	74.6	80.1	82.2	84.2	85.6	86.4	84.8	81.2	76.1	69.3
	5%	WB	57.7	59.2	61.1	64.9	68.9	74.2	76.4	77.3	75.0	71.5	67.2	61.0
		MCDB	64.5	66.9	71.0	76.8	80.6	83.0	84.5	85.2	83.3	79.6	73.8	67.3
	10%	WB	56.2	57.6	59.7	63.3	67.7	73.1	75.5	76.5	73.9	70.3	65.3	59.5
		MCDB	62.7	64.9	68.6	74.2	78.8	81.7	83.6	84.4	82.7	78.5	72.1	65.9
Mean Daily Temperature Range	5% DB	MDBR	17.3	18.5	19.4	20.7	20.0	17.5	14.8	15.0	16.5	17.7	17.3	16.0
		MCDBR	20.9	24.9	28.2	30.9	28.8	23.6	16.7	16.0	19.8	20.3	20.1	18.8
		MCWBR	11.5	12.3	12.0	11.6	9.5	8.1	6.4	6.0	7.6	8.7	9.0	10.1
	5% WB	MCDBR	19.0	20.9	23.0	25.4	22.7	18.3	15.4	15.4	17.3	18.1	17.8	16.9
		MCWBR	11.5	11.9	11.5	11.4	9.0	7.5	6.5	6.2	7.5	8.2	8.9	10.3
Clear-Sky Solar Irradiance	taub	0.394	0.437	0.464	0.509	0.502	0.435	0.422	0.428	0.437	0.431	0.400	0.391	
	taud	2.187	2.047	1.977	1.883	1.935	2.168	2.218	2.205	2.177	2.185	2.245	2.225	
	Ebn at Noon	258	257	259	252	255	271	274	272	264	256	254	252	
	Edh at Noon	38	49	56	63	61	48	45	45	45	42	36	36	
All-Sky Solar Radiation	RadAvg	1050	1325	1770	2140	2383	2559	2503	2291	1977	1530	1145	953	
	RadStd	46	80	59	69	71	33	21	22	29	45	54	37	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (4 neighbors)	+0.99	N/A	-1.17	+0.95	+0.67	N/A	N/A	-104	+358	+254

Nomenclature: See separate page